Improper Integrals

Questions 1–39

Question 1. Explain why each of the following are improper integrals.

a) $\displaystyle\int_1^\infty \sqrt{x}\,dx$

b) $\displaystyle\int_0^1 \frac{1}{3x^3}\,dx$

Question 2. Showing the limiting process used, find:

a) $\displaystyle\int_2^\infty \frac{1}{x^2}\,dx$

b) $\displaystyle\int_2^\infty \frac{1}{x^3}\,dx$

c) $\displaystyle\int_2^\infty \frac{1}{x^4}\,dx$

Question 3. Showing the limiting process used, find:

a) $\int_0^2 \frac{1}{x^2}\,dx$

b) $\int_0^2 \frac{1}{x^3}\,dx$

c) $\int_0^2 \frac{1}{x^4}\,dx$

Question 4. Showing the limiting process used, find:

a) $\int_0^9 \frac{1}{\sqrt{x}}\,dx$

b) $\int_4^\infty \frac{1}{\sqrt{x}}\,dx$

Question 5. Showing the limiting process used, find:

a) $\displaystyle\int_0^8 \frac{1}{x^{3/4}}\,dx$

b) $\displaystyle\int_8^\infty \frac{1}{x^{3/4}}\,dx$

Question 6. Showing the limiting process used, find:

a) $\displaystyle\int_4^\infty 192x^{-4}\,dx$

b) $\displaystyle\int_4^\infty (192x^{-4}+1)\,dx$

c) $\displaystyle\int_4^\infty 192x^{-4}(x+1)\,dx$

Question 7. Show clearly that $\displaystyle\int_0^4 \frac{1}{x^3\sqrt{x}}\,dx$ diverges.

Question 8. Show that $\displaystyle\int_0^{1/64} \left(\frac{1}{\sqrt{x}} - \frac{1}{\sqrt[3]{x}}\right) dx = \frac{5}{32}$.

Question 9. Amelia is attempting to evaluate the integral $\int_{-2}^{2} \frac{35}{x^6}\,dx$. Amelia's working is shown below:

$$\int_{-2}^{2} \frac{35}{x^6}\,dx = \int_{-2}^{2} 35x^{-6}\,dx = [-7x^{-5}]_{-2}^{2} = ((-7 \times 2^{-5}) - (-7 \times (-2)^{-5})) = -\frac{7}{32} - \frac{7}{32} = -\frac{7}{16}.$$

a) Explain the error in Amelia's method and what she should do instead.

b) Show clearly that $\int_{-2}^{2} \frac{35}{x^6}\,dx$ diverges.

Question 10. Find $\int_{-1}^{8} \frac{4}{x^{2/3}}\,dx$.

Question 11. Find $\displaystyle\int_{-32}^{1} \frac{100}{x^{4/5}}\,dx$. **The following questions require A-Level Maths Integration methods, including the Integrals of Standard Functions, Reversing the Chain Rule, Integration by Substitution, Integration by Parts, and Partial Fractions.**

Question 12. Explain why each of the following are improper integrals.

a) $\displaystyle\int_{4}^{6} \frac{1}{x-5}\,dx$

b) $\displaystyle\int_{0}^{1} \frac{e^x}{e^x-2}\,dx$

c) $\displaystyle\int_{0}^{2\pi/3} \frac{\sqrt{3}\sin x+\cos x}{\sin x-\sqrt{3}\cos x}\,dx$

d) $\displaystyle\int_{0}^{1} x^4 \ln x\,dx$

Question 13. Showing the limiting process used, find:

a) $\displaystyle\int_2^\infty \frac{1}{x}\,dx$

b) $\displaystyle\int_0^2 \frac{1}{x}\,dx$

Question 14. Showing the limiting process used, find:

a) $\displaystyle\int_0^\infty \frac{1}{e^x}\,dx$

b) $\displaystyle\int_0^\infty \frac{1}{e^{2x}}\,dx$

Question 15. Show that $\displaystyle\int_7^\infty (x-3)^{-5/2}\,dx = \frac{1}{12}$.

Question 16. Showing the limiting process used, find $\displaystyle\int_{-2}^{62.5} \frac{2}{(2x+3)^6}\,dx$.

Question 17. Showing the limiting process used, find:

a) $\displaystyle\int_3^4 \frac{x}{x^2-9}\,dx$

b) $\displaystyle\int_3^4 \frac{x}{\sqrt{x^2-9}}\,dx$

Question 18. Showing the limiting process used, find:

a) $\displaystyle\int_0^\infty 4xe^{-x^2}\,dx$

b) $\displaystyle\int_{-\infty}^\infty 4xe^{-x^2}\,dx$

Question 19. Showing the limiting process used, find $\displaystyle\int_{-\infty}^{0} \frac{e^x}{e^x+1}\,dx$.

Question 20. Showing the limiting process used, find $\displaystyle\int_{0}^{\pi/2} \tan x\,dx$.

Question 21. Showing the limiting process used, find $\displaystyle\int_{-1}^{2} \frac{x+1}{\sqrt{8-2x-x^2}}\,dx$.

Question 22.

a) Show that $\displaystyle\lim_{a\to\infty}\left(\frac{6a-5}{3a+7}\right) = 2$.

b) Showing the limiting process used, find $\displaystyle\int_{4}^{\infty}\left(\frac{6}{6x-5} - \frac{3}{3x+7}\right)\,dx$.

Question 23. Show that $\displaystyle\int_1^{\infty} \frac{2}{x(3x+1)}\,dx = \ln k$, where k is a rational number to be found.

Question 24. Showing the limiting process used, find $\displaystyle\int_{-1}^{2} \frac{2-x}{\sqrt{x+1}}\,dx$.

Question 25. Showing the limiting process used, find $\displaystyle\int_{1/\pi}^{\infty} \frac{\sin(1/x)}{x^2}\,dx$.

Question 26. Using the substitution $u = 2 - x^3e^{-2x}$ and showing the limiting process used, show that

$$\int_{3/2}^{\infty} \frac{(3-2x)x^2}{x^3 - 2e^{2x}}\,dx = \ln\left(\frac{16e^3}{16e^3 - 27}\right).$$

The following questions require A-Level Further Maths Integration methods, including Inverse Trigonometric Functions, Hyperbolic Functions and Partial Fractions.

Question 27. By showing the limiting process used, show that $\int_{-1}^{1} \frac{1}{\sqrt{1-x^2}}\,dx = \pi$.

Question 28. Showing the limiting process used, find $\int_{0}^{\infty} (\cosh x - \sinh x)\,dx$.

Question 29. Showing the limiting process used, find:

a) $\displaystyle\int_2^\infty \frac{1}{4x^2-9}\,dx$

b) $\displaystyle\int_0^\infty \frac{1}{4x^2+9}\,dx$

Question 30. By showing the limiting process used, show that

$$\int_0^\infty \frac{12-6x}{(2x+3)(x^2+3)}\,dx = \ln k,$$

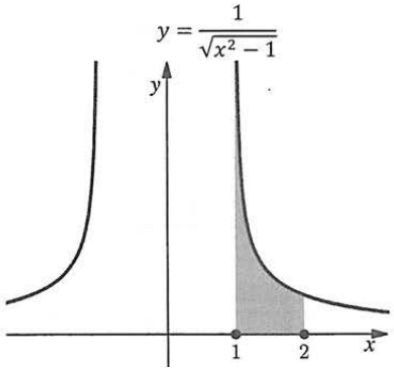
where k is a rational number to be found.

Question 31. By showing the limiting process used, show that

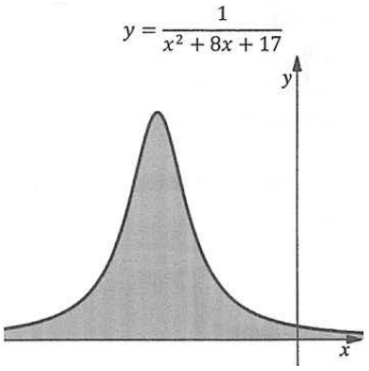
$$\int_2^\infty \frac{3x^2+16}{x^2(x^2+4)}\,dx = 2 - k\pi,$$

where k is a rational number to be found.

Question 32. The diagram shown below shows the curve $y = \dfrac{1}{\sqrt{x^2-1}}$. Find the exact shaded area between the curve, the x-axis, the line $x = 1$ and the line $x = 2$.



Question 33. The diagram shown below shows the curve $y = \dfrac{1}{x^2 + 8x + 17}$. Find the exact total shaded area between the curve and the x-axis.



Using L'Hôpital's Rule

Question 34. Show that $\displaystyle\int_0^1 \frac{\ln x}{\sqrt{x}}\,dx = -4$.

Question 35. Showing the limiting process used, find $\int_0^{12} \ln x \, dx$.

Question 36. Showing the limiting process used, find $\int_0^{\infty} x^2 e^{-3x} \, dx$.

Question 37.

a) Use L'Hôpital's Rule to find $\lim_{x\to\infty}\left(\frac{\ln x}{x}\right)$.

b) Showing the limiting process used, find $\int_1^\infty \frac{\ln x}{x^2}\,dx$.

Question 38.

a) Use L'Hôpital's Rule to find $\lim_{x\to\infty}\left(\frac{x}{e^x}\right)$.

b) Showing the limiting process used, find $\int_{-\infty}^0 xe^x\,dx$.

Question 39. Show clearly that $\displaystyle\int_3^\infty \frac{\ln x - 1}{(\ln x)^2}\,dx$ diverges.